

Bike Showroom Management System — Implementation Plan

1. Project Overview

A web + mobile application to manage inventory, GST-compliant billing, service history, worker accounts, attendance, and reporting for a single bike showroom, replacing Tally for showroom-specific operations.

Personas: Owner/Admin, Technician, Billing/Cashier, Delivery, Auditor (a staff account can hold multiple personas; Auditor stays isolated read-only)

2. Recommended Tech Stack

Layer	Recommendation	Why
Backend	Node.js (NestJS) or Django (Python)	Both handle RBAC, REST/GraphQL APIs, and GST calc logic cleanly
Database	PostgreSQL	Strong relational integrity for inventory/ledger data, good reporting support
Web frontend	React (Next.js)	Fast, component-driven, good ecosystem for dashboards
Mobile	React Native	Single codebase shared logic/components with web team
Auth	JWT + refresh tokens, role-based access control (RBAC)	Matches persona-based permission model
File/report export	PDF (invoices), Excel/CSV (GST, reports)	Matches GSTR upload + owner reporting needs
Hosting	Cloud VM or managed PaaS (e.g., AWS/GCP/Azure, or Railway/Render for lower cost)	Single showroom scale doesn't need complex infra initially

Notifications (later phase)	SMS/WhatsApp API for service-ready alerts	Optional enhancement
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3. Initial Setup / Onboarding

On first install, before any inventory or service data is entered, the app should walk the owner through a one-time **setup wizard** to configure showroom identity and business details. This becomes the source of truth used across invoices, GST filings, and reports.

Setup wizard fields:

- Showroom name, logo, address, contact number, email
- GSTIN (GST registration number), PAN
- State (for CGST/SGST vs IGST calculation on invoices)
- Invoice numbering format/prefix (e.g., `SHW-2026-0001`)
- Default GST rate(s) to pre-fill when adding inventory (can still be overridden per item)
- Currency/locale (in case of future multi-region use)
- Owner's own login credentials (first admin account)

Stored as: a single `ShowroomProfile` entity (one row, since this is single-showroom for now), referenced by the invoice/report templates so it doesn't need to be re-entered anywhere else.

Why this matters:

- GSTIN + state are required for correct invoice tax splitting (CGST/SGST for intra-state, IGST for inter-state)
- Showroom name/logo/address appear on every invoice and estimate — needs to be set once, correctly, before any billing happens
- Keeps the door open for multi-branch later — `ShowroomProfile` can become a table with multiple rows instead of a schema rewrite

4. High-Level Data Model

Core entities:

- `User` (id, name, phone, password_hash, personas[], is_active)
- `Attendance` (user_id, check_in, check_out, date)
- `InventoryItem` (id, name, category/type, hsn_code, gst_rate, unit_price, stock_qty, reorder_level)
- `Bike` (id, reg_no, chassis_no, model, customer_id)
- `Customer` (id, name, phone, address)
- `ServiceTicket` (id, bike_id, technician_id, complaint, estimate_amount, actual_amount, status: intake/approved/in-service/billed/delivered)
- `ServicePartUsed` (ticket_id, inventory_item_id, qty, price_at_use)
- `Estimate` (ticket_id, line_items[], gst_breakup, discount, created_by)
- `Invoice` (id, ticket_id or sale_id, gst_breakup, discount, total, cashier_id, created_at)
- `DelegationTask` (assigned_to, task_type, status: pending/approved/rejected, reviewed_by)
- `AuditLog` (actor_id, action, entity, before/after, timestamp)

Relationships to note:

- `ServiceTicket` drives inventory deduction (via `ServicePartUsed`)
- `Invoice` can originate from a `ServiceTicket` (service billing) or standalone sale
- `DelegationTask` + `AuditLog` together give the owner-review workflow and traceability

5. Role-Based Access Control (RBAC) Matrix

Action	Owner	Technician	Cashier	Delivery	Auditor
Configure GST rates	✓	✗	✗	✗	✗
Add/edit inventory	✓	✓ (if delegated)	✗	✗	view only
Search service logs	✓	✓ (any bike)	view only	view only	view only
Create estimate	✓	✓	✗	✗	✗

Approve delegated tasks	✓	✗	✗	✗	✗
Generate invoice	✓	✗	✓	✗	✗
Mark delivered	✓	✗	✗	✓	✗
View dashboard/reports	✓	✗	✗	✗	✓
Export GST/reports	✓	✗	✗	✗	✓
Manage worker accounts	✓	✗	✗	✗	✗
View attendance	✓	own only	own only	own only	✗

6. Tally Migration Strategy

1. **Export from Tally:** Stock items, stock groups, ledgers, GST rate masters, opening balances — via Tally's XML export.
2. **Build a mapping layer:** Tally stock groups → app's item categories; Tally ledger heads → app's chart of accounts equivalent (for reporting continuity).
3. **One-time import script:** Parses Tally XML/CSV, validates GST rates and HSN codes, loads into `InventoryItem` and opening stock/balance tables.
4. **Reconciliation step:** Generate a diff report (Tally totals vs. imported totals) for the owner/auditor to sign off before go-live.
5. **Parallel run (recommended):** Run both systems for 2–4 weeks before fully retiring Tally, to catch mismatches early.

7. Phased Build Roadmap

Phase 1 — Foundation (Weeks 1–3)

- First-run setup wizard: showroom name, GSTIN, address, invoice numbering, default GST rate

- User auth, RBAC/persona system, attendance check-in/out
- Inventory module: add/edit items, GST rate config, categories
- Basic web app shell + mobile app shell

Phase 2 — Service Workflow (Weeks 4–6)

- Bike/customer records, service ticket intake, estimate generation
- Service log search (global, any bike)
- Parts-used tracking with inventory deduction

Phase 3 — Billing & Delivery (Weeks 7–8)

- Cashier persona: invoice generation, GST breakup, discounts
- Delivery persona: delivery confirmation, ticket closure
- Delegation task queue + owner approval workflow

Phase 4 — Reporting & Migration (Weeks 9–10)

- Owner dashboard (sales trends, GST liability, technician performance)
- GST-ready export (GSTR-1 style, HSN-wise summary)
- Tally data migration tooling + reconciliation

Phase 5 — Mobile Parity & Hardening (Weeks 11–12)

- Full mobile feature parity for technician/cashier/delivery flows
- Audit log UI, edge-case testing, performance tuning
- Parallel run support alongside Tally

Phase 6 — Go-Live (Week 13+)

- Staff training, parallel run monitoring, cutover from Tally

Timeline assumes a small dev team (2–3 engineers); adjust based on actual team size.

8. Risks & Mitigations

Risk	
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	Mitigation
GST rate misconfiguration during migration	Mandatory reconciliation report before go-live
Staff resistance to new system	Parallel run period + simple mobile UX for daily tasks
Data loss during Tally export/import	Keep Tally read-only backup; test import on a copy first
Delegation misuse (worker over-editing)	All delegated actions logged and go through approval queue

9. Open Items for Next Discussion

- Multi-branch support — out of scope now, but worth designing DB schema to allow it later without a rewrite
- SMS/WhatsApp notifications for service status — nice-to-have, not core
- Payment gateway/UPI integration for cashier persona — confirm if needed at launch or later